

```
from google.colab import files
uploaded = files.upload()
```

Screenshot... 142745.png

Screenshot 2026-08-07 142745.png(image/png) - 94094 bytes, last modified: 8/7/2026 - 100% done
Saving Screenshot 2026-08-07 142745.png to Screenshot 2026-08-07 142745.png

```
import cv2
import matplotlib.pyplot as plt

# Read the image in grayscale
# The 'image.jpg' file does not exist in the Colab environment.
# Instead, use the name of the file that was uploaded via `files.upload()`.
# The variable 'uploaded_filename' holds this name.
img = cv2.imread(uploaded_filename, cv2.IMREAD_GRAYSCALE)

# Check if image was loaded successfully
if img is None:
    print(f"Error: Could not load image '{uploaded_filename}'. Please ensure it's a valid image.")
else:
    # Apply Sobel operator along X-axis
    sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)

    # Apply Sobel operator along Y-axis
    sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

    # Combine X and Y gradients
    sobel_xy = cv2.magnitude(sobel_x, sobel_y)

    # Convert to 8-bit image
    sobel_xy = cv2.convertScaleAbs(sobel_xy)

    # Display images
    plt.figure(figsize=(12, 5))
```

```
plt.subplot(1, 3, 1)
plt.imshow(img, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(cv2.convertScaleAbs(sobel_x), cmap="gray")
plt.title("Sobel X")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(sobel_xy, cmap="gray")
plt.title("Sobel XY Edge Detection")
plt.axis("off")

plt.show()
```

Original Image



Sobel X



Sobel XY Edge Detection



